

Presentation: Monotone maps, sphericity and bounded second eigenvalue by Bilu, Linial

CSE 269

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Key results

- δn regular graphs with bounded diameter D have sphericity $\Omega(\frac{n}{\lambda_2+1})$ where λ_2 is the second largest eigenvalue of the adjacency matrix
- quasi-random graphs have logarithmic sphericity
- $\frac{n}{2}$ regular graphs w/ bounded λ_2 are $o(n^2)$ close to bipartite graphs

Definition (sphericity)

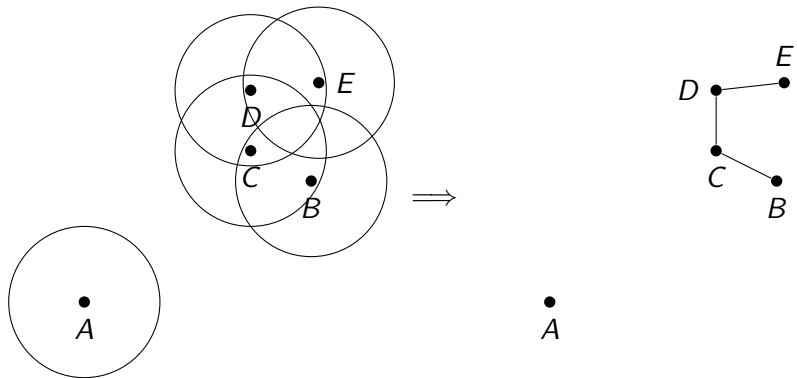
given a graph G and adjacency matrix A , its sphericity is the lowest dimension n ST $\varpi : G \rightarrow \mathbb{R}^n$ satisfies

$$A_{ij} = 1 \iff |\varphi(i) - \varphi(j)|_2 \leq 1 \quad (1)$$

- this map would be an isometry if we used the metric on the original graph of

$$\begin{aligned} \delta'(i, j) &= 0 \iff i = j \\ \delta'(i, j) &= 1 \iff \delta(i, j) = 1 \\ \delta'(i, j) &= 2, \text{ else} \end{aligned} \quad (2)$$

Sphericity



(3)

- use linear algebra/spectral theory on adjacency matrices for info
- n regular graphs have largest eigenvalue of n
- second eigenvalue gives good info on “variance” of edges by expander mixing lemma for some subset $S, T \subset G$

$$\left| E(S, T) - \frac{d}{n}|S||T| \right| \leq \lambda_2 \sqrt{|S||T|} \quad (4)$$

- λ_2 controls for “worst case deviation”
- good expander graphs lack local clusters; local clusters are easy to compress

Theorem Statement

Theorem

Let G be a d -regular graph with diameter D and λ_2 be the second largest eigenvalue of G 's adjacency matrix and at least $d - \frac{1}{2}n$. then

$$Sph(G) = \Omega\left(\frac{d - \lambda_2}{D^2(\lambda_2 + O(1))}\right) = \Omega\left(\frac{n}{\lambda_2 + 1}\right)$$

a couple observations:

- if X is real symmetric,

$$\text{rank}(X) \geq \frac{(\text{tr}(x))^2}{\sum_{i,j} X_{ij}^2} \quad (5)$$

- if X is PSD, then $\exists$ a set v_i such that $X_{ij} = v_i \cdot v_j$. X is also called a Gramian matrix
- if the embedding of an enumerated graph $w_i \in V$ is given by

$$\varphi : w_i \mapsto v_i \quad (6)$$

for vectors v_i comprising of the gramian matrix X , then the dimension of the embedding $= \text{rank}(X)$

Idea: what we will do is construct a feasible gramian matrix where the rank is $\frac{d-\lambda_2}{D^2(\lambda_2+O(1))} \approx \frac{n}{\lambda_2+1}$

- X : $n \times n$ symmetric matrix
- J : all ones matrix
-

$$R : (i, j) \in \mathbb{R}^{n \times n} \mapsto X_{ij} \quad (7)$$

$$R(X) \in \mathbb{R}^{n \times n}$$

rows are all the same element, i th row is just all X_{ij}

- $Y: 2X - R(X) - R(X)^T + J$

- why is Y like that?

- $Y_{ii} = 1$



$$\begin{aligned} Y_{ij} &= 2\langle x_i, x_j \rangle - \langle x_i, x_i \rangle - \langle x_j, x_j \rangle + 1 \\ &= 1 - |x_i - x_j|^2 \end{aligned} \tag{8}$$

- applying rank inequality 5

$$\text{rank}(Y) \geq \frac{n^2}{n + \sum_{i \neq j} (1 - |x_i - x_j|^2)^2} \tag{9}$$

- new goal: show

$$\sum_{i \neq j} (1 - |x_i - x_j|^2)^2 = O(D^2 n^2 \frac{\lambda_2}{d - \lambda_2}) \tag{10}$$

- we can normalize by D^2 . now all $|x_i - x_j| \approx 1$.
- separate ij pairs into two groups
 - ij is an edge,
 - ij is not an edge

feasible embedding $\implies$

$$\begin{cases} |v_i - v_j| \leq 1 & \iff (i, j) \in E(G) \\ |v_i - v_j| > 1 & \iff (i, j) \notin E(G) \end{cases} \quad (11)$$

- max eq 10 $\iff$

$$\max \sum_{ij \notin E} (|x_i - x_j|^2 - 1) + \sum_{ij \in E} (1 - |x_i - x_j|^2) \quad (12)$$

$$\begin{aligned}
 \max \quad & \sum_{(i,j) \notin E} (V_{ii} + V_{jj} - 2V_{ij}) + \sum_{(i,j) \in E} (-V_{ii} - V_{jj} + 2V_{ij}) - \binom{n}{2} + nd \\
 \text{s.t.} \quad & V \in PSD \\
 & \forall (i,j) \in E \quad V_{ii} + V_{jj} - 2V_{ij} \leq 1 \\
 & \forall (i,j) \notin E \quad V_{ii} + V_{jj} - 2V_{ij} \geq 1
 \end{aligned}$$

The dual problem is:

$$\begin{aligned}
 \min \quad & \frac{1}{2} \text{tr} A \\
 \text{s.t.} \quad & A \in PSD \\
 & \forall (i,j) \in E \quad A_{ij} \leq -1 \\
 & \forall (i,j) \notin E, i \neq j \quad A_{ij} \geq 1 \\
 & \forall i \in [n] \quad \sum_{j=1, \dots, n} A_{ij} = 0
 \end{aligned}$$

Create matrix that satisfies conditions for A

- the matrix $A = (\alpha d - n)I + J - \alpha M$ where M is the adjacency matrix satisfies all of these conditions if $\alpha > 2$.
- on edges, $A_{ij} = 1 - \alpha$, on non edges $A_{ij} = 1$ and we want edge A_{ij} to be less than -1 by our optimization problems, thus $\alpha \geq 2$
- finally, we can check that A 's eigenvalues are the same as M 's, with an eigenvector u having eigenvalue $(\alpha d - n) - \alpha \lambda_i$. to ensure this is psd, alpha is set to $\frac{n}{d - \lambda_2}$.
 - note that $A\vec{1} = 0$
- This A gives an upper bound on the value 12

$$\begin{aligned}\frac{1}{2}trA &= \frac{1}{2}n(\alpha d - n + 1) \\ &= \frac{1}{2} \left(n^2 \frac{d}{d - \lambda_2} - n^2 + n \right) \\ &= \frac{1}{2} \left(n^2 \frac{\lambda_2}{d - \lambda_2} + n \right)\end{aligned}\tag{13}$$

we have finished what we set out to prove in eq 10



some results on bipartite graphs

$$Sph(K_{m,n}) \leq m + \frac{n}{2} - 1 \quad (14)$$

$$Sph(K_{n,n}) \geq n \quad (15)$$